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A single, fair 6-sided die is tossed. Find the odds in favor of rolling a 1, 2, 3, or 4.

The odds in favor of an event E are given by the ratio $\frac{P(E)}{P(E')}$.

When a die is rolled, the outcome will be a number between 1 and 6 inclusive. All of the numbers have an equal probability of occurring.

Let E be the event that a number rolled is 1, 2, 3, or 4. Count the number of outcomes in which event E occurs and divide this by the total number of possible outcomes.

$$P(E) = \frac{\text{number of outcomes in E}}{\text{total possible outcomes}} = \frac{4}{6}$$

Then E' is the event that a number rolled is *not* 1, 2, 3, or 4. Remember the complement rule: for any event E, $P(E') = 1 - P(E)$. Subtract the probability of E from 1.

$$P(E') = 1 - \frac{4}{6} = \frac{2}{6}$$

The odds in favor of rolling a 1, 2, 3, or 4 are given by $\frac{P(E)}{P(E')} = \frac{\frac{4}{6}}{\frac{2}{6}}$. Simplify this ratio.

$$\frac{P(E)}{P(E')} = \frac{4}{2} = \frac{2}{1}$$

Therefore, the odds in favor of rolling a 1, 2, 3, or 4 are $\frac{2}{1}$, written 2 to 1.